第六次作业

习题五 1(4)(5) 2(2)(7)(8) 3(2)(8)(9)

5 8(3) 9(2) 11(1)

1(4) $\cos\frac{1}{z}=1-\frac{1}{2!}(\frac{1}{z})^2+\frac{1}{4!}(\frac{1}{z})^4+\cdots$

故 $z\cos\frac{1}{z}=z-\frac{1}{2!}\frac{1}{z}+\frac{1}{4!}(\frac{1}{z})^3+\cdots$

$z=0$ 为 $z\cos\frac{1}{z}$ 的本性奇点

(5) $\frac{z^2}{\cosh z-1}=\frac{z^2}{\frac{z^2}{2}+\frac{z^4}{4!}+\cdots}$

$z=0$ 是 z^2 的二级零点，是 $\cosh z-1$ 的零点

故 $z=0$ 是 $\frac{z^2}{\cosh z-1}$ 的可去奇点

2. (2) $\tan z=\frac{\sin z}{\cos z}$

$\cos z=0 \Rightarrow z=\frac{\pi}{2}+k\pi$，为孤立单极点

故 $\operatorname{Res}[\tan z, \frac{\pi}{2}R]=-1$

(7) $z=-1$ 为 $\sin\frac{z}{z+1}=\sin(1-\frac{1}{z+1})=\sin 1\cos\frac{1}{z+1}-\cos 1\sin\frac{1}{z+1}$

$=\sin 1-\cos 1\frac{1}{z+1}-\sin 1\frac{1}{2!(z+1)^2}+\cos 1\frac{1}{3!(z+1)^3}-\cdots$

故 $\operatorname{Res}[\sin\frac{z}{z+1}, -1]=-\cos 1$

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(8) $z=1$，设 $z+1=t$，则 $z^2e^{\frac{z+1}{z-1}}=(t+1)^2e^{\frac{t}{t}}$

$=(t+1)^2+\frac{(t+1)^2}{t}+\frac{(t+1)^2}{2t^2}+\frac{(t+1)^2}{6t^3}+\cdots$，$t$ 系数为 $\frac{13}{6}$

故 $\operatorname{Res}[z^2e^{\frac{z+1}{z-1}};1]=\frac{13}{6}$

3. (2) $\oint_C \frac{z}{(z-1)(z-2)^2}dz$ $C:|z-2|=\frac{1}{2}$

$z=2$，为 $\frac{z}{(z-1)(z-2)^2}$ 二级极点

原式 $=2\pi i\operatorname{Res}[\frac{z}{(z-1)(z-2)^2};2]$

$=2\pi i\lim_{z\to 2}\frac{-1}{(z-1)^2}=-2\pi i$

(8) $\oint_{|z|=1}\frac{\cos(e^{-z})}{z^2}dz$

$z=0$，为 $\frac{\cos(e^{-z})}{z^2}$ 二级极点

原式 $=2\pi i\operatorname{Res}[\frac{\cos(e^{-z})}{z^2};0]$

$=2\pi i\ e^{-z}\sin(e^{-z})\big|_{z=0}=2\pi i\sin 1$

(9) $z_k=k\pi$，$k=-2,-1,0,1,2$

原式 $=2\pi i\sum\operatorname{Res}[\cot z;z_k]=10\pi i$

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$$\text{设 } f_1(z)=\frac{\varphi_1(z)}{z-z_0},\ f_2(z)=\frac{\varphi_2(z)}{z-z_0}$$

5、z_0是$f_1(z)\cdot f_2(z)$的单极点$\Leftrightarrow z_0$是$\frac{1}{f_1(z)}$,$\frac{1}{f_2(z)}$的单零点

故z_0是$\frac{1}{f_1(z)f_2(z)}$的二级零点$\Leftrightarrow z_0$是$f_1(z)f_2(z)$的二级极点,

$$\operatorname{Res}[f_1(z)\cdot f_2(z);z_0]=\lim_{z\to z_0}\left[(z-z_0)^2\frac{\varphi_1(z)}{z-z_0}\frac{\varphi_2(z)}{z-z_0}\right]'$$

$$=\varphi_1'(z_0)\varphi_2(z_0)+\varphi_1(z_0)\varphi_2'(z_0)$$

$$=\varphi_1'(z_0)\operatorname{Res}[f_2(z);z_0]+\varphi_2'(z_0)\operatorname{Res}[f_1(z);z_0]$$

8(3). 设$z=e^{i\theta}$ 则$dz=ie^{i\theta}d\theta$

$$d\theta=\frac{1}{ie^{i\theta}}dz$$

$$\text{原式}=\oint_{|z|=1}\frac{1}{1-2\rho\frac{z+\frac{1}{z}}{2}+\rho^2}\frac{1}{iz}dz$$

$$=\oint_{|z|=1}\frac{-i}{(-\rho z+1)(z-\rho)}dz$$

$$=2\pi i\operatorname{Res}\left[\frac{-i}{(-\rho z+1)(z-\rho)};\rho\right]$$

$$=\frac{2\pi}{1-\rho^2}$$

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10(2) $\int_0^\infty \frac{x^2}{(x^2+a^2)^2}dx$

$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)^2}dx$

$= \frac{1}{2} 2\pi i \cdot \text{Res}\left[\frac{x^2}{(x^2+a^2)^2}; ia\right]$

$= \pi i \lim_{x\to ia}\left[\frac{x^2}{(x+ia)^2}\right]' =$

$= \pi i \lim_{x\to ia} \frac{2xia}{(x+ia)^3} = \frac{\pi}{4a}$

11(1) $\int_0^\infty \frac{x\sin x}{1+x^2}dx$

$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{x\sin x}{1+x^2}dx$

$= \text{Im}\left(\frac{1}{2}\int_{-\infty}^{\infty} \frac{xe^{ix}}{1+x^2}dx\right)$

$= \text{Im}\left\{\pi i \,\text{Res}\left[\frac{xe^{ix}}{1+x^2}; i\right]\right\}$

$= \text{Im}\left\{\frac{\pi i e^{-1}}{2}\right\}$

$= \frac{\pi}{2e}$

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复变函数与积分变换习题本

作业6.补

思考题交：2.7(理由)，8

习题交：1，4(5)(6)

思考题交

补：充分性：取G为单位圆，$w_0=0$，$\alpha=0$，即得证

必要性：取单位圆域$D_1=\{z:|z|<1\}$

取$z_0\in D$，$0\in D_1$，$\alpha=0$，存在$D\to D_1$的保角双射，$f(z_0)=0$，$\arg f'(z_0)=0$

取$w_0\in G$，$0\in D_1$，$\alpha\in(-2,2]$，存在$G\to D_1$的保角双射，$g(w_0)=0$，$\arg g'(w_0)=\alpha$

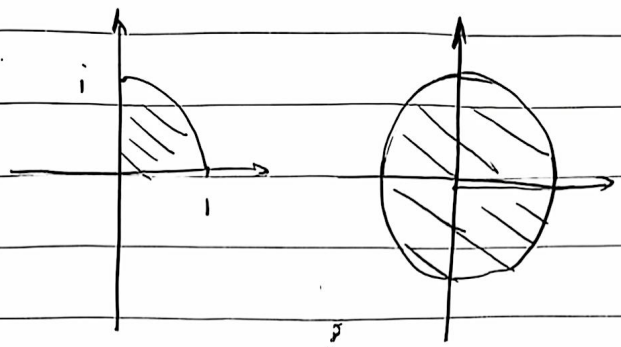
又任取$z_0\in D$，$w_0\in G$，$\alpha\in(-2,2]$，

必存在$D\to G$的保角双射$w=g^{-1}[f(z)]$

$$\frac{dw}{dz}=\frac{1}{\frac{dg}{dw}}f'(z)\Rightarrow f'(z_0)>0,\ \arg\left(\frac{dw}{dz}\Big|_{z=z_0}\right)=-\arg g'(w_0)=\alpha$$

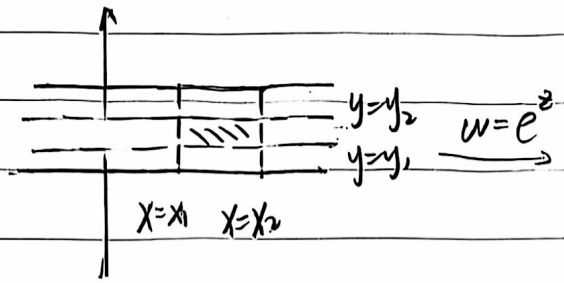
故$w_0=g^{-1}[f(z_0)]$，$\arg[g^{-1}f(z)]'|_{z=z_0}=\alpha$，即得证

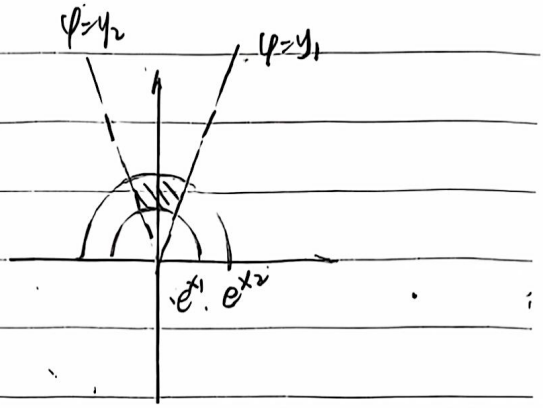
T7



不是，是单位圆内挖去[0,1]

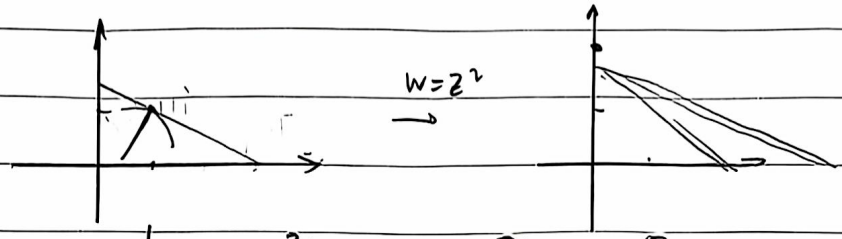
T8





习题六

T1

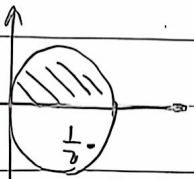


$\arg w'|_{z=1+i} = \frac{2}{4}$ 故 $\beta = \frac{2}{6} + \frac{2}{4} = \frac{5}{12}\pi$

T4 (5) $z=\frac{1}{w} \Rightarrow \begin{cases} x=\frac{u}{u^2+v^2} \\ y=-\frac{v}{u^2+v^2} \end{cases}$ 故 $x^2+y^2+2x+2y+1=0$，化简

$\Rightarrow u^2+v^2+2u-2v+1=0$

T6 $z=\frac{1}{w}=\frac{1}{u+iv}=\frac{u-iv}{u^2+v^2}$



故 $\begin{cases} x=\frac{u}{u^2+v^2} \\ y=\frac{v}{u^2+v^2} \end{cases} \Rightarrow \begin{cases} v>0 \\ 0<\frac{v}{u^2+v^2}<1 \end{cases} \Rightarrow \begin{cases} v>0 \\ u>0 \\ u^2+v^2-v>0 \end{cases}$，为半圆

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